

Correlating two computations

Prediction and classification attacks between a reference computation and an encoded one

2026-07-27

A tutorial: the general problem of correlating two computations, the three families of predictor that attack it, and what each of our protection plans does against each attack. Measurements are on the $n=128$ sliced sandwich with the Gray-code fold (docs/GRAY_FOLD.CG).

1 The general setting

Take two computations run on the same inputs: C (the reference — what we are trying to hide) and C' (what an adversary holds). Fix a prefix i of C and a prefix j of C' , run both on s shared random inputs $x_1 \dots x_s$, and you have two binary matrices

$$Y = C_i, \quad Y[a, b] = (\text{state of } C \text{ after } i \text{ gates on } x_a)_b \quad (s \times n)$$

$$Z = C'_j, \quad Z[a, b] = (\text{state of } C' \text{ after } j \text{ gates on } x_a)_b \quad (s \times m)$$

Every attack below tests whether Z tells you about Y , evaluated at each (i, j) and read as a heatmap. In our case $C' = G$, the gadgetized circuit, with $m = 4n$ wires against C 's n ; the extra inputs are pinned to zero.

2 The trivial answer, and why security must be class-relative

Ask “are Y and Z dependent?” and the answer is **always yes**. Both are deterministic functions of the same x , and G is an invertible encoding, so Z determines x determines Y . The mutual information is maximal by construction.

This dictates what can be measured. A test that is *consistent against all alternatives* — distance correlation, HSIC, any universal independence test — fires at full strength on every build, secure or not. **No test can certify “no dependence.”** Every meaningful statement has the form

*no adversary in class **A** can extract feature **f** of C 's state from G 's state, at sample budget s ,*

so every number below is a **lower bound**, and the class must always be named. That is not modesty; it is the only claim the setting admits.

3 The unifying object: the cross-correlation spectrum

Fix i, j . For a target mask $\alpha \in \mathbb{F}_2^n$ and a predictor mask $\beta \in \mathbb{F}_2^m$, define

$$M[\alpha, \beta] = \mathbb{E}_x \left[(-1)^{\alpha \cdot Y \oplus \beta \cdot Z} \right].$$

This is the **Walsh cross-spectrum**. $|M| = 1$ means $\alpha \cdot Y$ is an exact affine function of Z ; $M = 0$ means that parity carries nothing; in between are biases. Every instrument is a restriction of this one object:

restriction	gives you
$ M = 1$ exactly, β unrestricted	algebraic / exact predictors (§4)
$ M $ maximized, β of bounded weight	statistical predictors (§5)
replace $\beta \cdot Z$ by any degree- $\leq D$ function of Z	degree- D versions of both
the full singular spectrum, over $\mathbb{R}$	canonical correlation (§6)

The matrix is $2^n \times 2^m$, so nobody enumerates it; every instrument picks a restriction, and its blind spots follow from that choice.

4 Algebraic (exact) predictors

The question. Is $\alpha \cdot Y$ *exactly* a degree- $\leq D$ function of Z , on every input?

How it is answered. Build the design matrix of all monomials of Z up to degree D — $N_D = \sum_{k \leq D} \binom{m}{k}$ columns — and test whether $\alpha \cdot Y$ lies in its GF(2) column span, with a held-out split to reject spurious fits. The answer is *binary per cell*.

Our instrument. `hmap_affine`, reporting $H(i, j) = \text{mean over } C\text{'s } n \text{ target bits of the holdout error, scoring an inconsistent bit at exactly 0.5. So } H \approx 0 \text{ is total leakage, } H \approx 0.5 \text{ is hidden.}$

Why it exists. It is *invariant to the affine part of the encoding*. G carries each value as a XOR of shares over unrelated wires; a raw Hamming comparison sees noise, and an early “the diagonal is destroyed” conclusion was exactly that artifact, corrected once the affine-invariant measure existed. Degree 2 additionally sees through any quadratic re-encoding.

Its blind spot. It cannot see *bias*. A relation holding on 99% of inputs scores the same as one holding on 50% — both are “not exact.” This is why it and the statistical measures disagree, and why neither subsumes the other.

5 Statistical (bias) predictors

The question. How well can a *bounded* predictor **guess** $\alpha \cdot Y$ from Z ?

Our instrument. `stress_battery`, with two families:

- **F1** — best over {constant, single wire, XOR of two wires}: $|\beta| \leq 2$, degree 1. A GF(2) analogue of correlation power analysis.
- **F2** — F1 plus one AND over a declared wire set (for us the 256 band wires): degree 2.

Three design points make the numbers readable:

1. **Lift, not agreement.** Each cell reports best – base rate, the base rate being the best *constant* predictor, so a merely skewed target scores zero.
2. **An overfit floor.** With $W = 512$ predictor wires the search is over $\sim 131\text{k}$ candidates, and maximizing over that many buys agreement by chance. The floor is the same search against a *random* target; **prominence** = raw – floor, and the leak gate is prominence $> 2 \cdot$ floor.
3. **Two gates.** ALIGNED-LEAK needs exploitable information *and* progress alignment (ρ , perm- z): the best-predicting G -prefix must advance as the C -prefix does.

Scaling. The floor behaves as $\sqrt{\ln N/2s}$, so resolving a bias δ needs $s \gtrsim \ln N/\delta^2$, while

$$\text{cost} = \text{rows} \times \text{cols} \times n_t \times \left(\frac{W^2}{2} + 2A^2 \right) \times \frac{s}{64}$$

is *linear* in s and *quadratic* in wires. Resolving a leak $k\times$ smaller costs k^2 ; widening G from $4n$ to $8n$ needs only $\sim 12\%$ more samples but $\sim 4.5\times$ the time.

6 Canonical correlation, and the right basis

Classical CCA maximizes $\text{corr}(Ya, Zb)$ over *real* vectors a, b , solved by an SVD of the whitened cross-covariance; the singular values are the canonical correlations. The translation to our setting is the useful part. Encode bits as ± 1 ; then

XOR of a set of bits $\longleftrightarrow$ PRODUCT of the ± 1 variables $\longleftrightarrow$ one Fourier character.

So a GF(2)-linear relation is a *multiplicative* monomial, not an additive one. Consequently:

- **Real-linear CCA** on raw bits finds *additive* combinations — essentially Hamming-weight relations. That is the classical side-channel leakage model, natural when the observable is analog.
- **GF(2) character analysis** finds XOR relations. That is *linear cryptanalysis*, natural for a Boolean circuit.

Both are canonical correlation over different bases. For our problem the XOR basis is the right one, and real CCA on raw wire values would mostly measure Hamming-weight structure the encoding neither especially protects nor exposes. The GF(2) analogue of CCA’s full singular spectrum is the entire Walsh cross-spectrum of §3 — which is why practical instruments are all *restrictions* rather than decompositions.

7 Our case: G is a masking scheme, so order matters

The product-share encoding represents each value as

$$v = w \oplus A_1 \oplus \dots \oplus A_k, \quad A_r = \text{product of } d_r \text{ band bits,}$$

with w the carrier. This is **Boolean masking**, and the side-channel literature on masked implementations transfers directly.

The piling-up lemma. For a degree- d product of uniform independent bits, $\mathbb{E}[(-1)^A] = 1 - 2^{1-d}$: a degree-2 atom contributes 0.5, a degree-3 atom 0.75. If the atoms are variable-disjoint (which

`draw_slot` enforces per value) the biases multiply:

$$\varepsilon = \prod_r (1 - 2^{1-d_r}), \quad \text{corr}(\text{carrier}, \text{value}) = \varepsilon,$$

so a predictor reading the bare carrier guesses the value with agreement $(1 + \varepsilon)/2$. ε is the single number governing the statistical attack.

Order of attack. To *remove* an atom of degree d the adversary must compute a degree- d monomial of band wires — a d -th order combining function. After cancelling a set S the residual is $\prod_{r \notin S}$. Hence F1 sees the full ε ; F2 can cancel one degree-2 atom and see $\varepsilon/0.5 = 2\varepsilon$; an F3 would cancel a degree-3 atom and see $\varepsilon/0.75$.

The design tension, in one line. A degree-2 atom is a *stronger* statistical masker (0.5 vs 0.75) but a *weaker* algebraic one — it lies inside a degree-2 adversary’s span, where a degree-3 atom does not. Low degree buys statistics; high degree buys algebra.

8 Results: each plan against each attack

Five mask plans, all with the Gray fold, $n=128$, same source C and sandwich, $s = 8192$, plate geometry matched per build.

8.1 Predicted bias, before and after cancellation

plan	atoms	ε (order 1)	after one 2-AND	after one 3-AND
[2,3,3] (<i>shipped</i>)	d2, d3, d3	0.28125	0.5625	0.375
[2,2,3]	d2, d2, d3	0.1875	0.375	0.25
[2,2,3,3]	d2, d2, d3, d3	0.140625	0.28125	0.1875
[2,2,2,3]	d2×3, d3	0.09375	0.1875	0.125
[2,2,2,3,3]	d2×3, d3, d3	0.0703	0.140625	0.09375

8.2 Measured

plan	ε	gates	vs base	reachable	F1 raw	F2 raw	verdict F1/F2
[2,3,3]	0.28125	808,618	—	95.47%	0.0817	0.1111	LEAK / LEAK
[2,2,3]	0.1875	634,790	−21%	97.01%	0.0536	0.0782	flat / LEAK
[2,2,3,3]	0.1406	864,497	+6.9%	96.16%	0.0435	0.0660	flat / flat
[2,2,2,3]	0.09375	692,653	−14%	97.53%	0.0318	0.0487	flat / flat
[2,2,2,3,3]	0.0703	924,284	+14%	96.56%	0.0258	0.0409	flat / flat

“Raw” = prominence + floor; compare arms by raw, because the floor is a noisy single-sample estimate (§9).

8.3 The law

Regressing the measured signal on the predicted piling-up bias:

$$\text{F1}_{\text{raw}} = 0.262 \varepsilon + 0.007 \quad (R^2 = 0.996), \quad \text{F2}_{\text{raw}} = 0.330 \varepsilon + 0.018 \quad (R^2 = 0.998).$$

Five plans spanning a $4\times$ range of ε , and the statistical leak is **linear in the piling-up product** to within half a percent. This identifies the leak as the carrier’s *marginal mask bias* and nothing else.

Two readings follow. The **slope ratio** $0.330/0.262 = 1.26$ is the payoff of second order: theory says a *perfectly targeted* 2-AND doubles the effective bias (ratio 2.0), so the blind search over $\binom{256}{2}$ pairs realizes about a quarter of it — a targeted F2 using ledger knowledge would reach the full factor. The **intercepts** (0.007, 0.018) are residual overfit inflation the floor subtraction does not remove, a reminder that “flat” is a statement about a threshold, not about zero.

8.4 What the algebraic attack says, and why it is flat across the board

`hmap.affine`, matched geometry, reads every plate measured as dead: $\text{depthMed} = 0.0000$, $\text{depth} = 0.044\text{--}0.046$ against an “alive” calibration of ≈ 0.35 , and **zero genuine interior rows**.

plate	degree	depth	depthMed	ρ	interior leaks
wide fold [2,3,3]	1	0.0438	0.0000	0.017	0 / 88
Gray fold [2,3,3]	1	0.0436	0.0000	0.016	0 / 88
wide fold [2,3,3]	2	0.0455	0.0000	−0.042	0 / 88
Gray fold [2,3,3]	2	0.0439	0.0000	0.016	0 / 88
[2,2,3]	2	0.0439	0.0000	0.016	0 / 88
[2,2,2,3]	2	0.0444	0.0000	−0.047	0 / 88

The last two rows are the ones that mattered: [2,2,3] and [2,2,2,3] trade degree-3 atoms for degree-2 ones, so they are where a degree-2 exact adversary would surface first if the margin were thinner than theory says. It is not. ([2,2,3,3] and [2,2,2,3,3] were not run at degree 2; they carry *more* degree-3 material than the plans that passed, so they are covered a fortiori.)

That is forced, not lucky. Exact degree- D reconstruction requires *every* atom to lie in the degree- D span, i.e. $D \geq \max_r d_r$. All five plans keep at least one degree-3 atom, so **the algebraic verdict is identical for every plan** and they differ *only* statistically.

What does differ is **redundancy, not threshold**: [2,2,3] and [2,2,2,3] hold a single degree-3 atom, so if one is ever compromised the value drops into degree-2 range; the other three plans keep two. A pure [2,2,2] plan would have no algebraic margin at all — statistically the best of all, and dead at $D = 2$.

8.5 The Pareto surprise

[2,2,2,3] **dominates the shipped** [2,3,3] **on every axis measured at once**: 14% fewer gates, higher store-reachability (97.53% vs 95.47%), $3\times$ lower statistical leak, identical algebraic standing.

It is affordable because of the Gray fold. Under the wide fold a block emits $(1+k)^{\text{arity}}$ fragments, so a mask term is *multiplicative* — 3 to 4 terms costs +56%. Under the Gray fold the product part is a fixed ~ 9 gates regardless of k , and a term costs only its own gather (~ 1 gate at degree 2, ~ 4 at degree 3). Mask-plan cost becomes **additive**, and trading a degree-3 atom for degree-2 atoms makes the circuit *smaller*.

9 How to read these numbers without fooling yourself

Five traps, all of which bit during this work.

1. **Never read a heatmap by its mean** — it saturates near 0.5. Read the ridge.
2. **Never read ρ alone on a plate with ports.** Both axes have unencoded ends; cell (0,0) compares C 's input against G 's input wires and reads perfectly in *every* build. On such a plate ρ is two plateaus ranked against noise in a dead middle: two equally-dead artefacts once scored $\rho = 0.448$ and 0.019. Report **interior rows with prominence**. Our degree-1 plates report $\text{perm-}z = 4.91$, $p = 0.0003$ — and are dead.
3. **The overfit floor is a noisy single-sample estimate, and the verdict rides on it.** Measured floors are not monotone in s : 0.0454, 0.0205, 0.0242, 0.0136 at $s = 2048, 4096, 8192, 16384$. The *same circuit* scored `flat` at $s=2048$ and `ALIGNED-LEAK` at $s=4096$. Compare by raw.
4. **A fixed `--g-step` across circuits of different length is a confound.** $\text{cols} = |G|/\text{g_step}$ and `row-best` is a max over columns, so a longer circuit scores leakier for free. A +24%-gate arm ran 52 columns against 42 and looked 13% worse; matching geometry removed a third to a half of that.
5. **“Flat” means “below threshold at this s .”** Floor $\propto 1/\sqrt{s}$. `[2,2,2,3]`'s F1 raw of 0.0318 sits just above a 0.0239 floor with $\rho = 0.996$ — the diagonal is still there; it no longer clears the bar.

10 What we did not test

- **A structure-aware adversary.** The mask atoms are *readable off the gate list* — the fold's own gates name the band wires of each atom. Such an adversary does not search; it reads and cancels exactly, achieving the full order- d factor rather than the blind search's quarter. Strictly stronger than anything run here, and what rolling, re-sourcing and epochs exist to complicate.
- **Targeted F3**, cancelling a degree-3 atom ($\varepsilon/0.75$): trivial targeted, $\sim 2800\times$ F2's cost blind.
- **Differential correlation**, ΔC against ΔG under input differences. The battery's `traj` target is a *temporal* difference, not an input one.
- **Machine-learned predictors** over all $4n$ wires.
- **Universal independence tests** are deliberately absent; §2 says why.

11 What was changed as a result

`[2,2,2,3]` is now the **production default** (`ProdConfig::production_single`), replacing `[2,3,3]`, together with the Gray fold. This analysis is the reason: the plan is cheaper (692,653 gates against 808,618, -14%), more digestible (97.53% store-reachable against 95.47%), and leaks $3.2\times$ less (F1 raw 0.0318 against 0.0817, both battery probes going from `ALIGNED-LEAK` to `flat`), with identical algebraic standing — both dead at degree 2, measured.

`--prod-ladder-cap` stays retired: at this plan the Gray fold already reaches 97.53% without it, and the residual wide gates are mask-slot emissions rather than fold material.

[2,2,2,3,3] is the **natural step-up** (`--prod-k-hi 2`): $\varepsilon = 0.0703$, F1 raw 0.0258 — the lowest measured — and *both* degree-3 atoms retained, for 924,284 gates (+14% against the old default rather than −14%). Take it when redundancy matters more than size: the default holds a single degree-3 atom, and if that one atom is ever compromised the value falls into degree-2 exact range with nothing behind it.

12 Practical summary

- The statistical leak is the carrier’s marginal mask bias, **linear in the piling-up product** ($R^2 \approx 0.996$). To reduce it, reduce ε .
- Refresh **rate** and refresh **source** do not touch it — measured flat across `--prod-refill-data 0/50/100`. They change the band’s provenance, not its marginal, and the marginal is what F1 sees.
- Low-degree-heavy plans win statistically at *every* attack order; degree-3 atoms buy the algebraic margin. Keep at least two for redundancy.
- [2,2,2,3] is Pareto-better than the shipped [2,3,3] on size, digestibility and statistical leak, with identical algebraic standing but one fewer degree-3 atom of margin. [2,2,2,3,3] buys that margin back for +14% gates and the lowest leak measured.